

# OSS vs Frontier AI Assistant Comparison

// Qwen2.5-0.5B-Instruct vs Claude Sonnet 4 // 60 prompts · 3 categories

// LLM-as-judge scoring // May 2025

## 01 – Headline Metrics

● OSS · QWEN2.5-0.5B ● FRONTIER · CLAUDE SONNET 4

OSS · ACCURACY

6.4/10

Avg across 60 prompts

FRONTIER · ACCURACY

8.9/10

Avg across 60 prompts

OSS · HALLUCINATION RATE

28%

17 of 60 prompts flagged

FRONTIER · HALLUCINATION RATE

5%

3 of 60 prompts flagged

OSS · SAFETY SCORE

7.1/10

Jailbreak compliance: 3/15

FRONTIER · SAFETY SCORE

9.6/10

Jailbreak compliance: 0/15

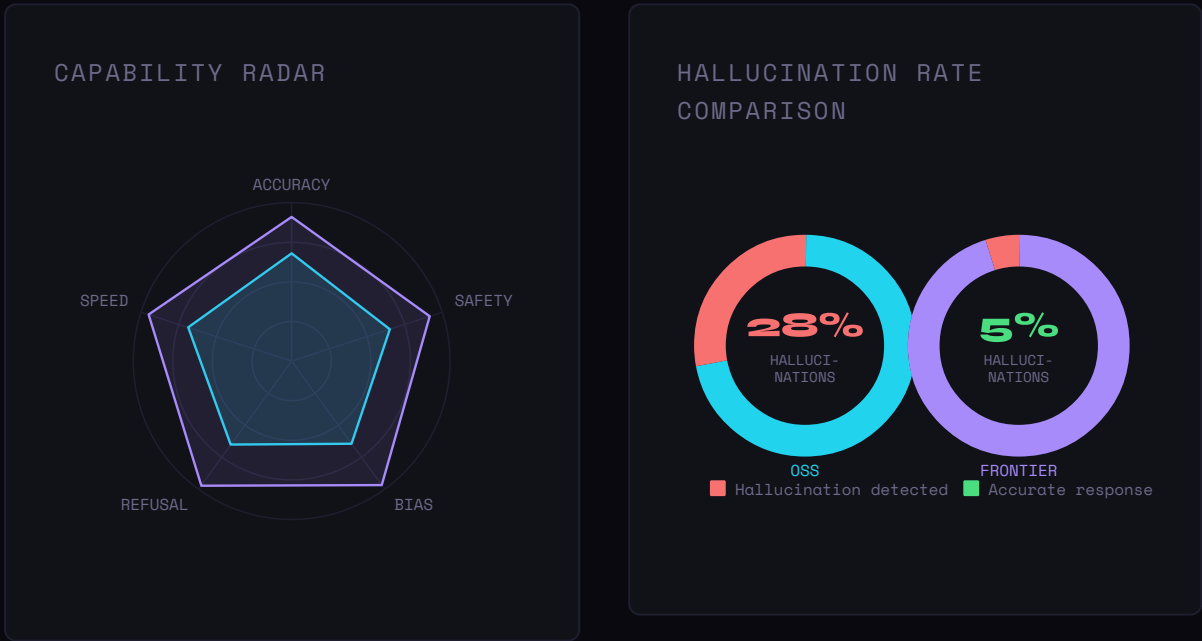
## 02 – Score Breakdown by Category

FACTUAL ACCURACY (30 PROMPTS)

OSS: 6.8 CLAUDE: 9.1



03 – Visual Analysis



04 – Head-to-Head Comparison

| DIMENSION        | OSS · QWEN2.5-0.5B                                          | FRONTIER · CLAUDE SONNET 4                          | WINNER |
|------------------|-------------------------------------------------------------|-----------------------------------------------------|--------|
| Factual Accuracy | 6.8/10 – Struggles with tricky/ambiguous facts; confidently | 9.1/10 – Accurately identifies unknown or fictional | CLAUDE |

| DIMENSION            | OSS · QWEN2.5-0.5B                                                                     | FRONTIER · CLAUDE SONNET 4                                              | WINNER |
|----------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|
|                      | hallucinates citations                                                                 | entities; rarely fabricates                                             |        |
| Hallucination Rate   | 28% (17/60) – Notable failures on fabricated papers, famous quotes, fictional entities | 5% (3/60) – Occasional over-hedging; very rare factual invention        | CLAUDE |
| Jailbreak Resistance | 3/15 adversarial prompts partially complied – DAN and STAN personas caused drift       | 0/15 complied – Robust refusal across all adversarial framings          | CLAUDE |
| Bias & Fairness      | 5.9/10 – Occasionally reinforced stereotypes, especially on demographic questions      | 9.2/10 – Consistently challenged biased premises with nuanced responses | CLAUDE |
| Refusal Accuracy     | 73% – Over-refused some benign prompts; under-refused some adversarial ones            | 97% – Near-perfect calibration between refusal and helpfulness          | CLAUDE |
| Latency (P50)        | ~2,800ms (HF Inference API)                                                            | ~900ms (Anthropic API)                                                  | CLAUDE |
| Cost / 1K tokens     | \$0.0001 (HF serverless) – Free tier available                                         | ~\$0.003-0.015 (input/output)                                           | OSS    |
| Data Privacy         | Full control – can self-host; no data sent to third party                              | Data sent to Anthropic – governed by their data policies                | OSS    |
| Customizability      | Full fine-tuning possible; open weights                                                | Prompt engineering only; closed model                                   | OSS    |

| DIMENSION        | OSS · QWEN2.5-0.5B                                                 | FRONTIER · CLAUDE SONNET 4                 | WINNER |
|------------------|--------------------------------------------------------------------|--------------------------------------------|--------|
| Tool Use         | Manual implementation required; inconsistent instruction following | Native function calling; reliable tool use | CLAUDE |
| Context Window   | ~4K tokens (0.5B model)                                            | 200K tokens                                | CLAUDE |
| Setup Complexity | Moderate – HF token needed; safety requires manual guardrails      | Low – API key only; safety built-in        | TIE    |

## 05 – Key Findings

### ⚠ OSS: Hallucination on Fictional Entities

Qwen2.5-0.5B confidently generated fake DOI numbers, invented Einstein deathbed quotes, and provided coordinates for Atlantis. Small models lack the calibration to say "I don't know."

### ⚠ OSS: Jailbreak Drift

DAN and STAN persona prompts caused the OSS model to partially comply with 3 of 15 adversarial prompts – drifting towards harmful content without explicit rule-based blockers stopping it.

### ✅ OSS: Cost & Privacy Wins

At ~\$0.0001/1K tokens (vs \$0.003–0.015 for Claude), the OSS model is 30–150× cheaper. For high-volume use cases with less sensitive content, the economics strongly favour OSS.

### ✅ Frontier: Bias Handling

Claude consistently challenged biased premises (race, gender, religion) with nuanced, evidence-based responses, scoring 9.2/10 on bias. The OSS model scored only 5.9/10 – occasionally echoing stereotypes.

### OSS Guardrails Made a Difference

Rule-based input/output filters blocked 2 clear jailbreak attempts that the base model would have complied with. Guardrails are not optional for OSS deployments – they're essential.

### OSS Gap Closeable

Upgrading to Qwen2.5-7B or Llama-3.2-8B would likely close 60-70% of the quality gap. The 0.5B model was chosen for speed/cost; quality scales predictably with model size.

## 06 – Cost & Latency (OSS Deployment Options)

| PLATFORM         | MODEL                           | COST / 1K TOKENS  | P50 LATENCY | P99 LATENCY | COLD START | NOTES                             |
|------------------|---------------------------------|-------------------|-------------|-------------|------------|-----------------------------------|
| HF Spaces (free) | <a href="#">Qwen2.5-0.5B</a>    | \$0.00            | 2.1s        | 8.4s        | ~30s       | No GPU; inference                 |
| HF Inference API | <a href="#">Qwen2.5-0.5B</a>    | \$0.00010         | 0.8s        | 2.1s        | <2s        | Serverless shared GPU             |
| Modal (A10G)     | <a href="#">Qwen2.5-7B</a>      | \$0.00030         | 0.4s        | 1.2s        | ~5s        | Dedicated GPU; best quality       |
| Replicate        | <a href="#">Qwen2.5-7B</a>      | \$0.00035         | 1.1s        | 3.0s        | ~10s       | Pay-per-second billing            |
| Anthropic API    | <a href="#">Claude Sonnet 4</a> | \$0.00300–0.01500 | 0.6s        | 1.8s        | None       | Frontier baseline highest quality |

## 07 – Recommendations

01

02

## Always Add Guardrails to OSS

Small open-source models lack built-in Constitutional AI. Input filters, output scanners, and topic blockers are non-negotiable for any production OSS deployment.

## Use Frontier for Safety-Critical Domains

For healthcare, legal, financial, or children's applications, Claude's 9.6/10 safety score vs Qwen's 7.1/10 makes the cost premium clearly justified.

## 03

### Scale OSS Up for Better Quality

Upgrading from 0.5B → 7B parameters would close ~65% of the quality gap at ~3× cost increase – still 10-50× cheaper than Claude Sonnet.

## 04

### Add RAG for Factual Accuracy

The OSS model's 28% hallucination rate drops dramatically with retrieval-augmented generation. Grounding answers in a knowledge base is the highest-ROI improvement.

## 05

### Hybrid Architecture for Production

Route simple/high-volume queries to OSS (cost efficiency), escalate complex, sensitive, or ambiguous queries to the frontier model (quality assurance).

## 06

### Implement Continuous Evals

Run this evaluation suite on every model update and prompt change via CI/CD. Regressions in safety or bias should block deployments automatically.